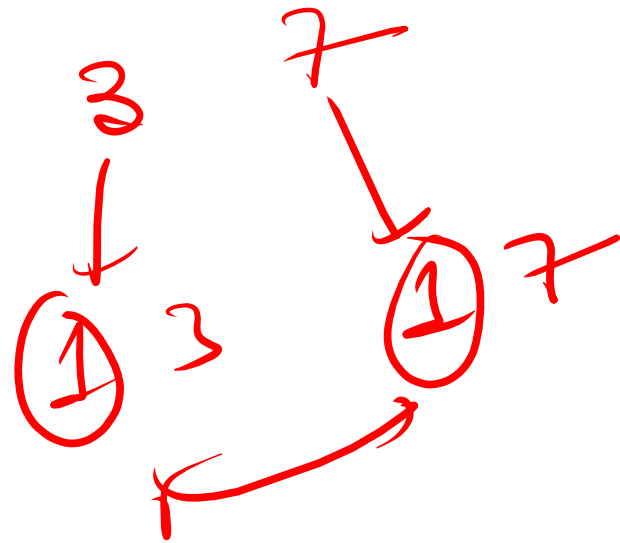
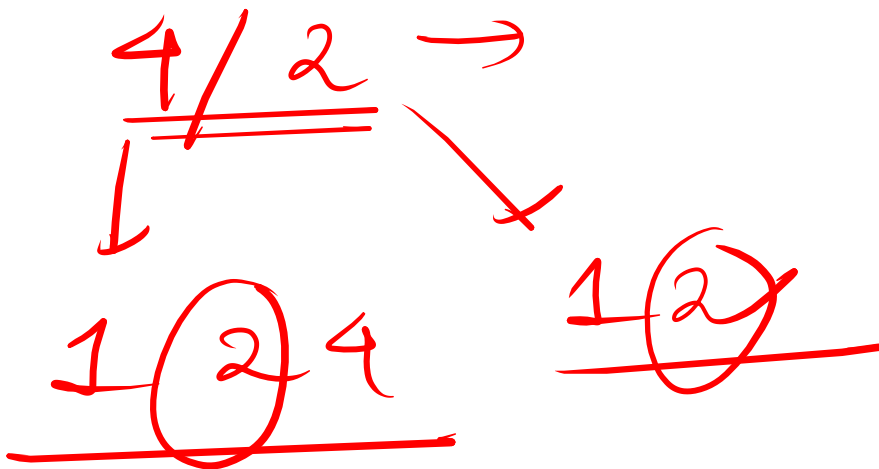
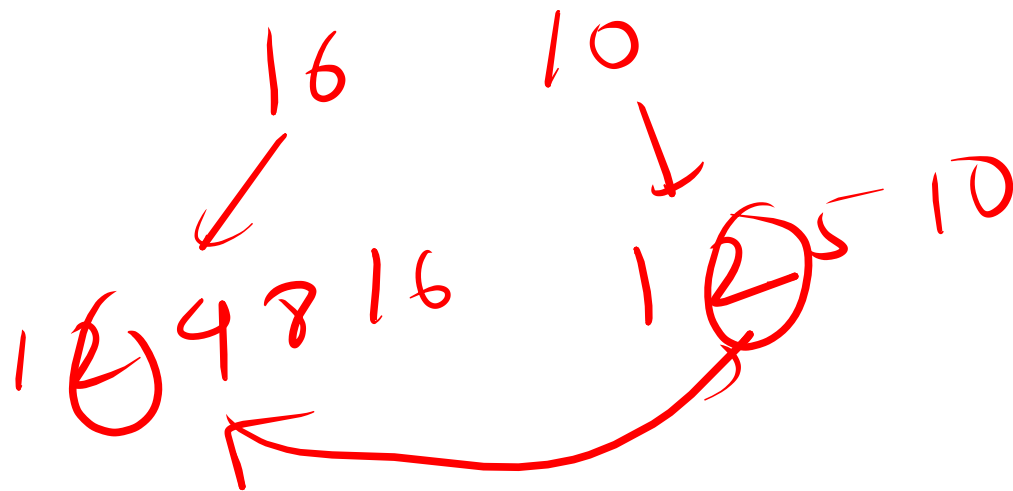


Greatest  
common  
div.

GCD



a b  
 ↓ ↓  
 common  
 div.  
 ↓  
 greatest



Euclidean theorem

$$\gcd(a, b) = \gcd(\overset{\downarrow}{b}, \overset{\downarrow}{a \div b})$$

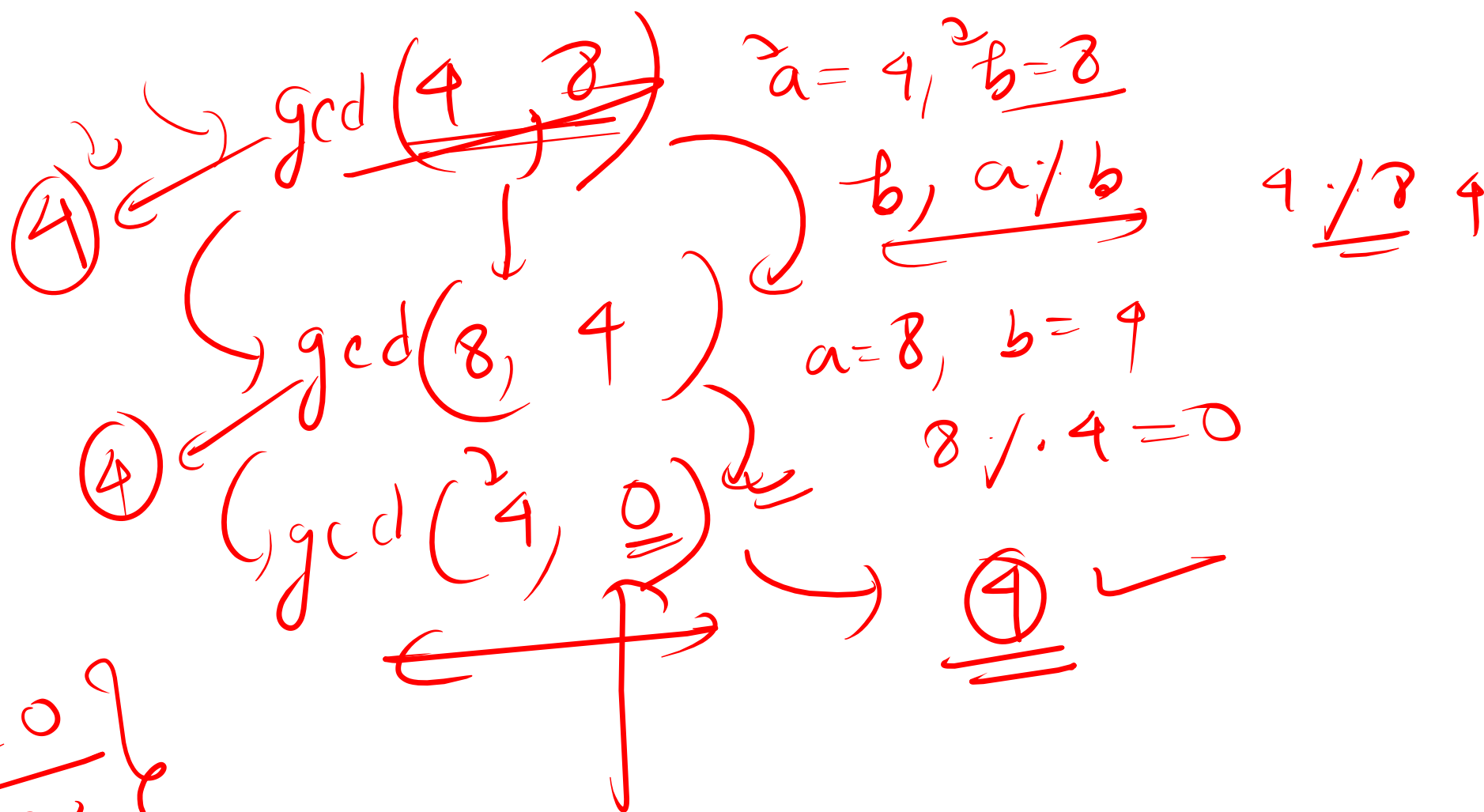
greater smaller

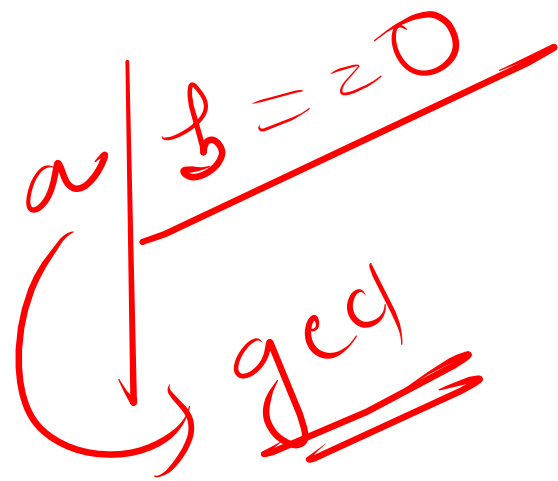
$\gcd(a, b)$

$\gcd(\underline{b}, \underline{a \div b})$

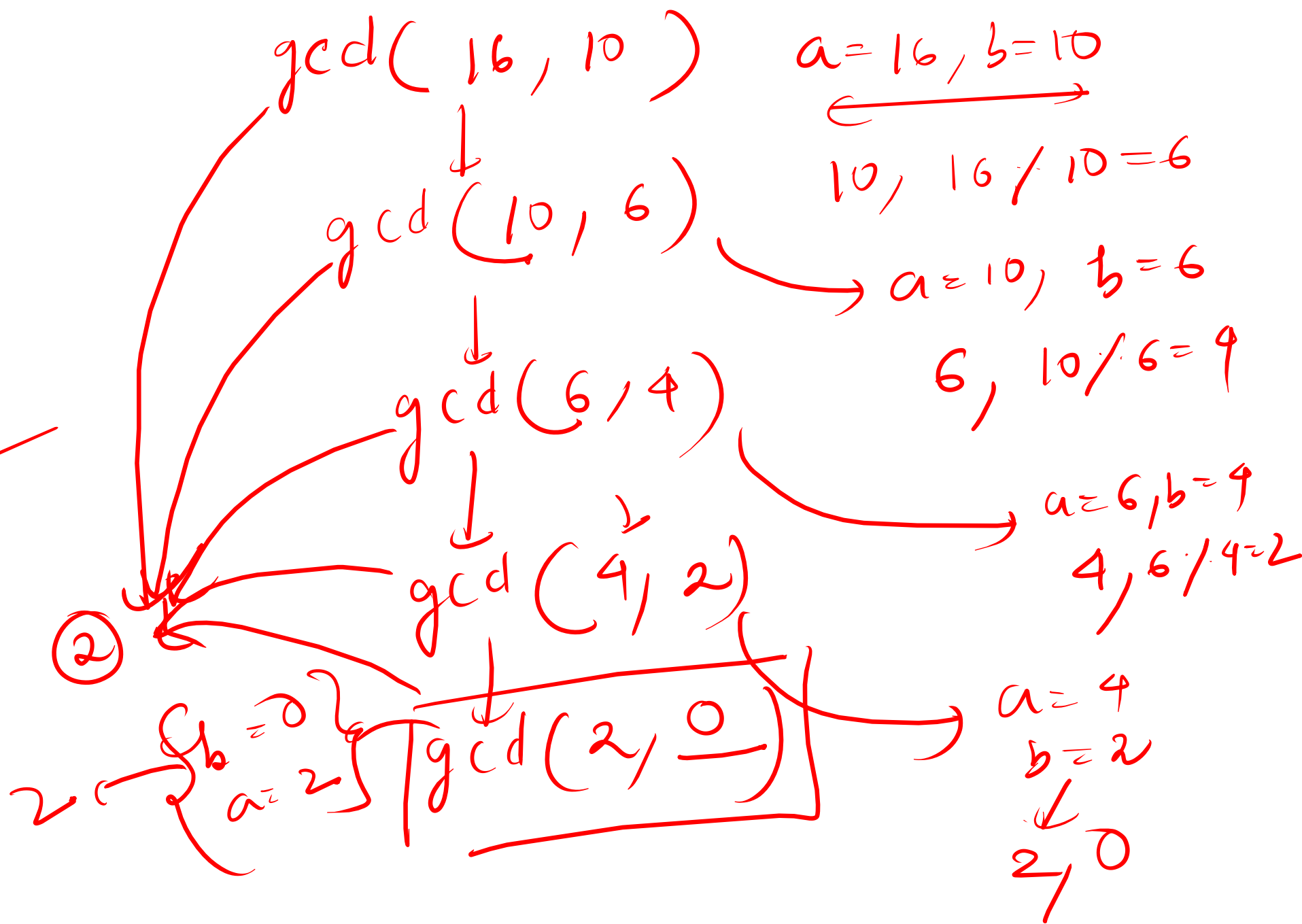


$$\left. \begin{array}{l} \frac{a}{a} \\ \frac{a}{a} \end{array} \right\} \quad \left. \begin{array}{l} \frac{b=0}{a} \end{array} \right\}$$





log(max(a/b))



LCM

$$6, 9 \rightarrow \underline{18}$$

$$\underline{3}, 7 \rightarrow 21 \rightarrow \frac{21}{1} = 21$$

$$2, 8 \rightarrow \underline{8} \rightarrow \frac{16}{2} = 8$$

$$4, 6 \rightarrow 12 \rightarrow \frac{24}{2} = 12 //$$

$$\frac{54}{3} = \textcircled{18}$$

$$\frac{a * b}{\gcd(a, b)}$$

$$\gcd \checkmark \rightarrow \text{LCM} = \underline{\underline{\frac{a * b}{\gcd(a, b)}}}$$

int  $\rightarrow a/b$

$\sim 10^6 * 10^6 \rightarrow$  long  
int  
consumed

long  
 $(10^6 / M) * (10^6 / M) / M$

$$(a+b) / \underline{M} \equiv (a/M + b/M) / M$$

$$(a-b) / M \equiv (a/M - b/M) / M$$

$$(a * b) / M \equiv ((a/M) * (b/M)) / M$$

$$M \approx 10^9 + 7$$

$$\underline{2^3} \rightarrow (2 * 2^2) / 7$$

$$\downarrow$$

$$(2 / 7 * (2^2) / 7) / 7$$

$$\downarrow$$

$$(2 * 4) / 7 = \textcircled{1}$$

$$\underline{2^5} / 7 \rightarrow 4$$

$$(2 * 2^4) / 7 = ((2 / 7) * (2^4) / 7) / 7$$

$$2^2 (2 * 2) / 7$$

$$\downarrow \downarrow$$

$$(2 / 7) * (2 / 7) / 7$$

$$\downarrow \downarrow$$

$$2 * 2 = \underline{4}$$

$$\downarrow$$

$$((2 / 7) * (2 * 2^3) / 7) / 7$$

$$\downarrow$$

$$(2 / 7) * (2 / 7) = \textcircled{4}$$

$$\downarrow$$

$$(2 / 7) * ((2 / 7) * (2^3) / 7) / 7$$

$$\downarrow \downarrow$$

$$\{ (2 * 1) / 7 \} = \underline{\textcircled{2}}$$

$$\rightarrow \textcircled{4}$$

$$(2^5)$$

mod  $\rightarrow$  i th power

$$P = (1 * 2) / 7 = ((1 / 7) * (2 / 7)) / 7 = \underline{\underline{2}}$$

$$(2 * 2) / 7$$

$$(2 / 7 * 2 / 7) / 7$$

$$(2 * 2) / 7 \rightarrow ((2 / 7) * (2 / 7)) / 7 = 4$$

$$(4 * 2) / 7 = ((4 / 7) * (2 / 7)) / 7 = 1$$

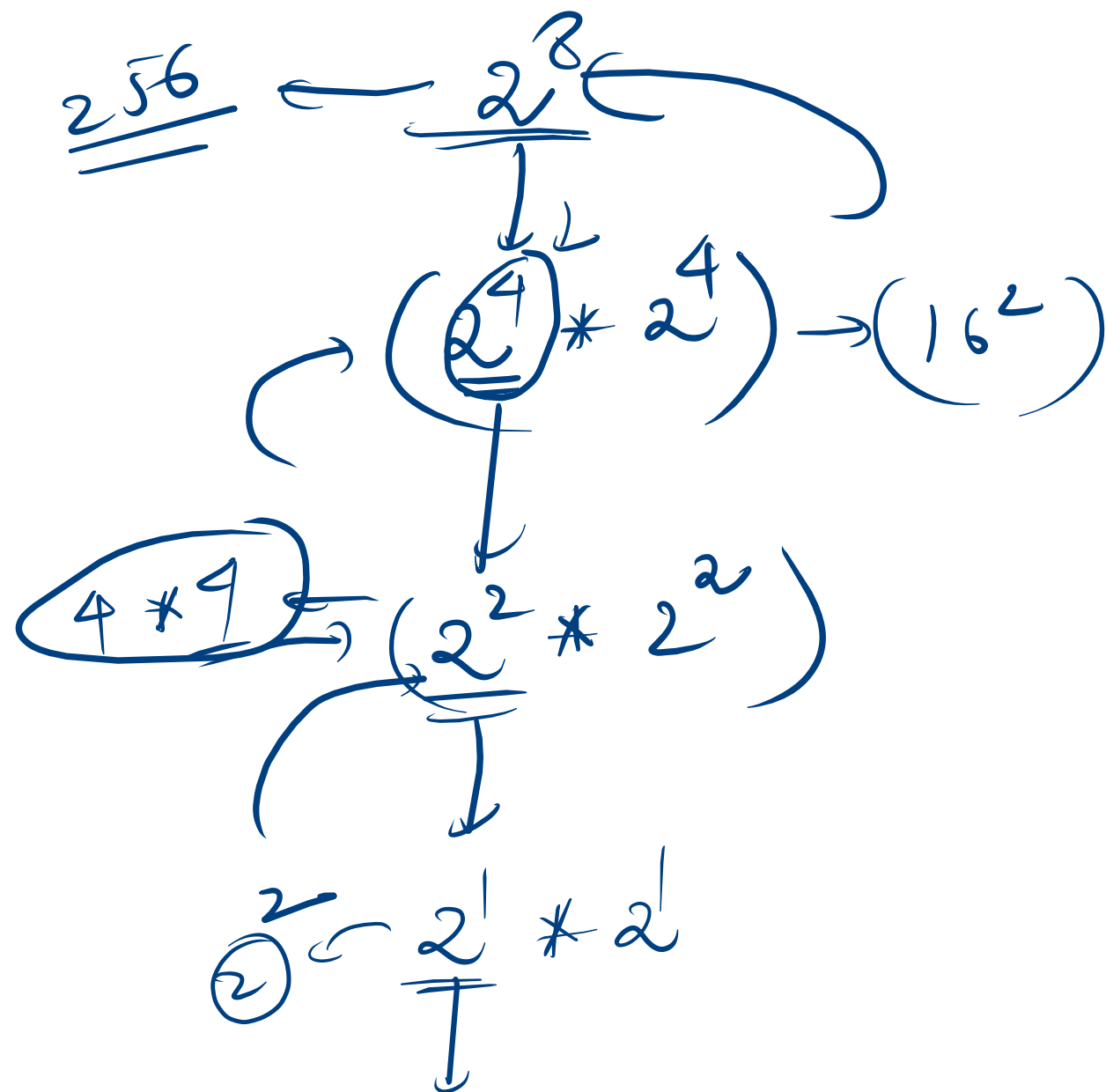
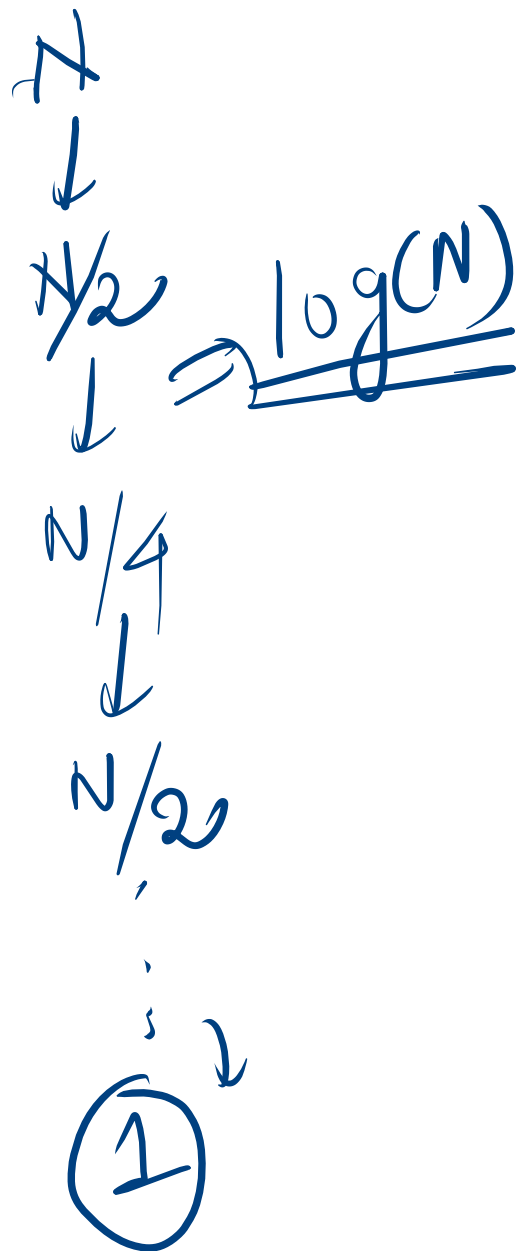
$$P \rightarrow \underline{\underline{4}}$$

$$(1 * 2) / 7 = ((1 / 7) * (2 / 7)) / 7 = \underline{\underline{2}}$$





⊕



$$\sqrt{6} \rightarrow \underbrace{110}$$

$$2^{\underline{6}} \rightarrow 2^{(2^2+2^1)}$$

$$2^{\textcircled{6}} \rightarrow \underline{110} \rightarrow \textcircled{2} + \underline{2} + \underline{0 \times 2^0}$$

|   |   |   |       |
|---|---|---|-------|
| 2 | 6 |   | 110   |
| 2 | 3 | 0 | $2^0$ |
| 2 | 1 | 1 | $2^1$ |
| 2 | 0 | 1 | $2^2$ |

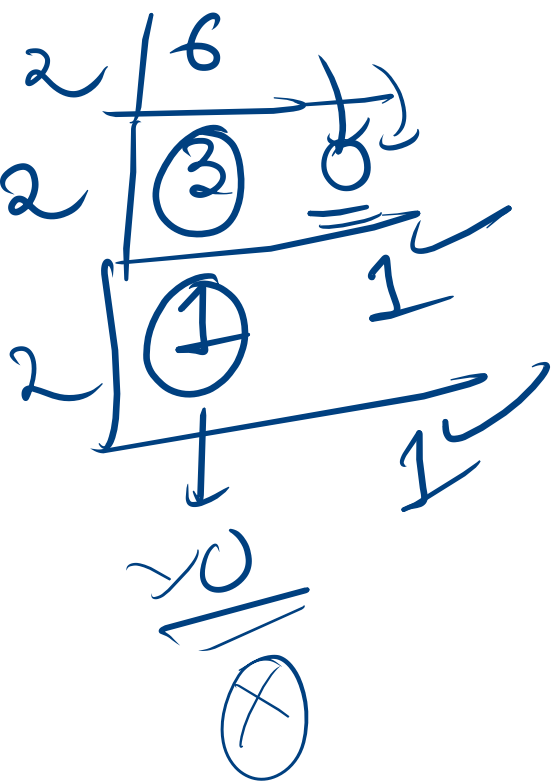
$$\log_2(N)$$

$$N \cdot \log_2(2)$$

$$\textcircled{2^x} \rightarrow \textcircled{N} = \textcircled{N}$$

$$2^{(4+2+0)} = 2^4 * 2^1 * 2^0$$

$$\text{rem } p \rightarrow \underline{\underline{2^{\text{exp}}}}$$



$$2^1 \boxed{P}$$

$$(2^2) \rightarrow N$$

$$2^{2^2} = 4$$

$$(2^4) \rightarrow$$

$$\underline{\underline{Pr.}}$$

$$(2 * 9) / 7$$

$$(1) \rightarrow$$

$$1 \rightarrow (2^P) \rightarrow N$$

$$(Pr * 2^P) / M$$

$$2^{\boxed{6}^2}$$

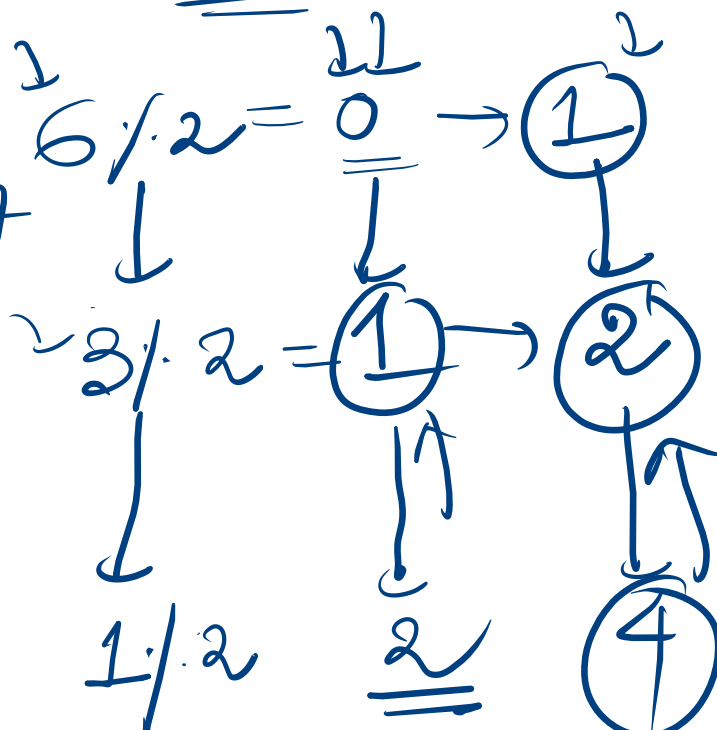
$$69 / 7 = 1$$

$$\underline{\underline{Pr = 1}}$$

$$((1/7)(2/7))/7$$

$$= 2$$

$$M = 7$$



$$1/2$$

$$(1)$$

$$\underline{\underline{2}}$$

$$3^{2^2+2^1+2^0}$$

$$\leftarrow \underline{\underline{7}}$$

$$3^4 + 2 + \underline{\underline{1}}^2$$

$$\downarrow \downarrow \downarrow$$

$$(\underline{\underline{3}})^9 * 3 * \underline{\underline{1}}$$

$$P \rightarrow (1 * 3) / 5$$

$$(3 * 9) / 5$$

$$(9 * 81) / 5 = \underline{\underline{4}}$$

$$7 \leftarrow \underline{\underline{2}}$$

$$N \log(N)$$

$$3 \dots \underline{\underline{1}}$$

$$\underline{\underline{M=5}}$$

$$M=1 \quad N=7/2=1 \rightarrow P=3$$

$$(1 * 1) / 5 = 3 \quad 3/2 = 1 \quad 1 * 3^2$$

$$(3 * 9) / 5 \quad 1/2 = 1$$

$$(3 * 4) / 5 = \underline{\underline{2}}$$

$$(2 * 81) / 5$$

$$(2 * \underline{\underline{1}}) / 5 = \underline{\underline{2}}$$

demo.  $\rightarrow \begin{cases} \text{non} \\ \text{prime} \end{cases}$

$$(a * x) \text{ / } n = 1$$

$$\Rightarrow x = \frac{a^{-1} \text{ / } n}{n \rightarrow \text{prime}}$$

$M \rightarrow \underline{\text{prime}}$

$\downarrow$

Fermat's little theorem

$$\left( \underline{a^p} - a \right) \text{ / } \underline{p} = \underline{0}$$

$\downarrow$

$$a^p \equiv a \text{ / } p$$

$$\hookrightarrow \underline{a^{p-1}} = 1 \text{ / } p \Rightarrow$$

$\uparrow$

$$\underline{\underline{a^{-1} \text{ / } p}}$$

$$(a^{p-1}) \pmod p \equiv (1 \pmod p)$$

$$(a^{p-2}) \pmod p = \overrightarrow{(a^{-1}) \pmod p}$$

$$(a^{-1}) \pmod p \equiv (a^{p-2}) \pmod p$$

$p \rightarrow \text{prime}$

$$\begin{aligned}
 & \underline{\underline{a^{-p+p-1}}} = \left( \bar{a}^{-p} \right) / \cdot p \\
 & \quad \quad \quad \swarrow \\
 & a^{-1} = \left( \bar{a}^{-p} \right) / \cdot p \\
 & \quad \quad \quad \underline{\underline{\hspace{10em}}}
 \end{aligned}$$



denominator  
 $\downarrow$   
prime

$$(a^p - a) / p = \underline{0}$$

<sup>2</sup>  
 { modular  
 exp }

$$(a^p) / p \equiv a / p$$

$$(a^p - a)$$

$$a * x \equiv 1 / p$$

$$x \equiv \left( \frac{1}{a} \right) / p$$

$$\circlearrowleft \textcircled{x} \leftarrow (a^{-1}) / p$$

$$a / p \equiv (a^p) / p$$

$$\times a^{-2} \quad \times a^{-2}$$

$$(a^{-1}) / p \equiv (a^{p-2}) / p$$

$\xrightarrow{\text{prime/non prime}}$   
 $\xrightarrow{\text{gcd}(a,b)}$   
 $\underline{\underline{x = (a^{-1}) / m}}$   
 $(n, m)$   
 $(nx) / m = 1 / m$   
 $\underline{\underline{x = (1/n) / m}}$

Extended ~~euclid~~  $\rightarrow$  gcd

$\downarrow$   
 $\hookrightarrow ax + by = \underline{\underline{\text{gcd}(a, b)}}$   
 $\downarrow$

$nx + my = \text{gcd}(n, m)$   
 $\downarrow$

$\downarrow$   
 $(\underline{nx} + \textcircled{my}) / m = (\text{gcd}(n, m)) / m$   
 $\downarrow$   
 $(nx) / m = (\textcircled{\text{gcd}}) / \underline{\underline{m}}$

$$n - \left(\frac{n}{m}\right) * m$$

$$\underline{a}x + \underline{b}y$$

$$\rightarrow \underline{n} - (\underline{m})$$

$$\begin{aligned} & \underline{\underline{7/3}} \downarrow \\ & 7 - \left(\frac{7}{3}\right) * \underline{\underline{3}} \\ & \quad \downarrow 2 * 3 \\ & 7 - 6 = \underline{\underline{1}} \end{aligned}$$

mod  
inverse

$$\underline{n}x + \underline{m}y = \gcd(n, m)$$

$$\downarrow \gcd(m, n/m)$$

$$mx_1, (n/m)y_1$$

$$\underline{\underline{n}}x + \underline{\underline{m}}y = \underline{\underline{m}}x_1 + \underline{\underline{(n/m)}}y_1$$

$$\begin{aligned} & \left(\frac{n}{m}\right) * \underline{\underline{m}} \\ & \quad \downarrow \\ & \underline{\underline{m}} \end{aligned}$$

$$g = x_1 - (n/m) y_1$$

$$x = y_1$$

$$n\bar{x} + my = mx_1 + \left( n - \binom{n}{m} * m \right) y_1$$

$$= \overset{1}{mx_1} + \overset{2}{ny_1} - m \binom{n}{m} y_1$$

$$\rightarrow \underline{\underline{\bar{x}}} + my = \left[ m \left( x_1 - \binom{n}{m} y_1 \right) + n \underline{\underline{y}} \right]$$

$$\textcircled{1} \quad n/m$$

$$(nx + my) \div m = \gcd(n, m) \div m$$

$$\left( (nx) \div m + \cancel{my} \div \cancel{m} \right) \div m = (1 \div m)$$

$$(nx) \div m = 1 \div m$$

$$\textcircled{x} \div \left( \frac{1}{n} \right) \div m = \left( \frac{n^{-1}}{\underline{\underline{\quad}}} \right) \div m \underline{\underline{\quad}}$$